Seatwork 6.1: Functions

Functions

Course Code: CPE 007	**Program: Computer Engineering**
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1. Objective(s)

This activity aims to develop programming skills using functions.

2. Intended Learning Outcomes

2.1 Create a program in C++ using functions with its parameters and arguments.

3. Materials and Equipment

Internet connection
Personal Computer

4. Procedure

Direction:

1. Create a program in C++ that will choose if the number is even, odd or invalid. Use functions in the program.
2. Create a program in C++ that will convert a string with LOWER CASES into ALL UPPER CASES using functions.

5. Output

1.

CODE:

```
#include <iostream>
#include <string>
using namespace std;

string checkNumber(int num) {
    if (num < 0) {
        // Negative numbers are treated as invalid in this program
        return "Invalid";
    } else if (num % 2 == 0) {
        // % gives the remainder; if dividing by 2 leaves 0, the number is even
        return "Even";
    } else {
        // Any other non-negative number has a remainder of 1, so it is odd
        return "Odd";
    }
}

int main() {
    int number;   // variable that stores the user's input

    // Ask the user to enter a number
    cout << "Enter a number: ";

    // cin >> number fails if the input is not a whole number
    if (!(cin >> number)) {
        cout << "Result: Invalid" << endl;
        return 0;   // end the program early
    }
```

```
    // Function call
    // Argument: 'number' is passed to the parameter 'num' of checkNumber()
    // The returned string is stored in 'result'
    string result = checkNumber(number);

    // Display the result
    cout << "Result: " << result << endl;

    return 0;
}
```

OUTPUT:

```
Enter a number: 176762
Result: Even

--------------------------------
Process exited after 7.455 seconds with return value 0
Press any key to continue . . .
```

2.

CODE:

```
#include <iostream>
#include <string>
using namespace std;

string toUpperCase(string text) {
    // Loop through every character in the string, one at a time
    // text.length() gives the total number of characters
    for (int i = 0; i < text.length(); i++) {

        // Check if the current character is a lowercase letter (a to z)
        if (text[i] >= 'a' && text[i] <= 'z') {

            // In ASCII, 'a' = 97 and 'A' = 65, a difference of 32
            // Subtracting 32 changes a lowercase letter to its uppercase form
            text[i] = text[i] - 32;
        }
        // Characters that are not lowercase letters (spaces, numbers,
        // symbols, and capital letters) are left unchanged
    }

    // Send the converted string back to the place where the function was called
    return text;
}

int main() {
    string input;   // variable that stores the user's sentence
```

```
    // Ask the user to type a string
    cout << "Enter a string in lowercase: ";

    // getline() reads the whole line, including spaces
    getline(cin, input);

    // Function call
    // Argument: 'input' is passed to the parameter 'text' of toUpperCase()
    // The returned string is stored in 'upper'
    string upper = toUpperCase(input);

    // Display the converted string
    cout << "Uppercase: " << upper << endl;

    return 0;
}
```

OUTPUT:

```
Enter a string in lowercase: small ito tapos magiging big
Uppercase: SMALL ITO TAPOS MAGIGING BIG

--------------------------------
Process exited after 11.33 seconds with return value 0
Press any key to continue . . .
```

6. Conclusion

In this activity, I learned how to use functions in C++ to make my programs more organized and easier to understand. I now understand that a parameter is the variable in the function definition, while an argument is the actual value passed to it when the function is called. Creating the even/odd/invalid checker taught me how to use conditions and return values, and the uppercase converter showed me how to process a string character by character using a loop. Overall, this exercise helped me see how functions make code reusable and how a problem can be broken into smaller, manageable parts.